WEEK 1 – POWER BI TASK : Superstore Sales Data Import, Power Query & Data Cleaning

INTRODUCTION:

This project focuses on importing, cleaning, and transforming the Superstore Sales dataset using Microsoft Power BI. The main objective of Week 1 was to prepare the raw dataset for further analysis and dashboard development. Power Query was used to inspect the data, correct data types, clean categorical fields, and create useful date-related attributes from the Order Date column.

Objective:

Objectives of the Project

- Import the Superstore Sales CSV dataset into Power BI.
- Inspect the dataset and its columns.
- Clean and standardize categorical data.
- Correct inappropriate data types and errors.
- Create date-related attributes from the Order Date.
- Prepare a clean dataset for future visualization and analysis.

Tools & Technologies Used

Tool	**Purpose**
Microsoft Power BI	Data analysis and visualization
Power Query	Data cleaning and transformation
CSV Dataset	Source of Superstore Sales data

Dataset Description

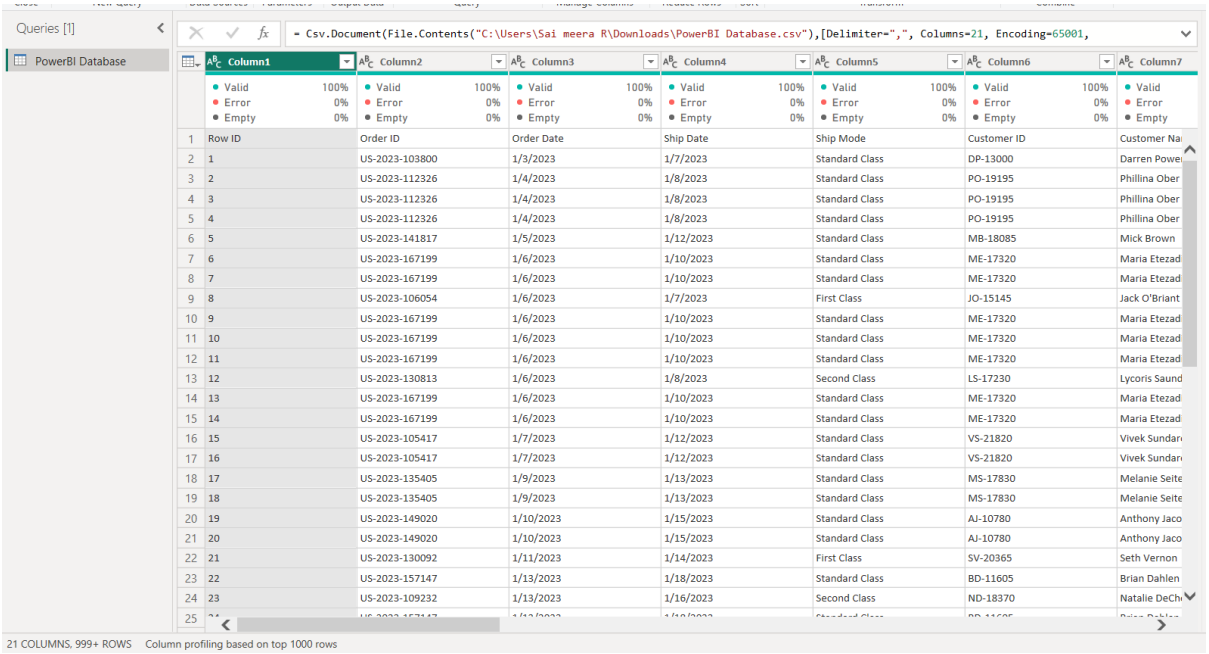
Dataset Used: Superstore Sales Dataset

The dataset contains sales-related information about customers, products, orders, sales, and other business attributes.

Some important columns include:

- Order ID
- Order Date
- Customer ID
- Category
- Sub-Category
- Segment
- Sales
- Quantity
- Discount
- Profit

Superstore Sales dataset imported into Power BI



Data Import:

The Superstore Sales CSV file was imported into Microsoft Power BI. After importing the dataset, the data was opened in Power Query Editor for inspection and transformation.

Data Cleaning & Transformation

Promoted Headers:

The first row of the dataset was promoted as the column headers so that the fields could be correctly identified.

Data Type Correction:

The data types of different columns were inspected and corrected where necessary.

One important correction was made to the Order ID column, which was incorrectly interpreted as a Date. It was changed to the appropriate Text data type.

Cleaning Categorical Columns:

The following categorical columns were cleaned and standardized:

- Category
- Sub-Category
- Segment

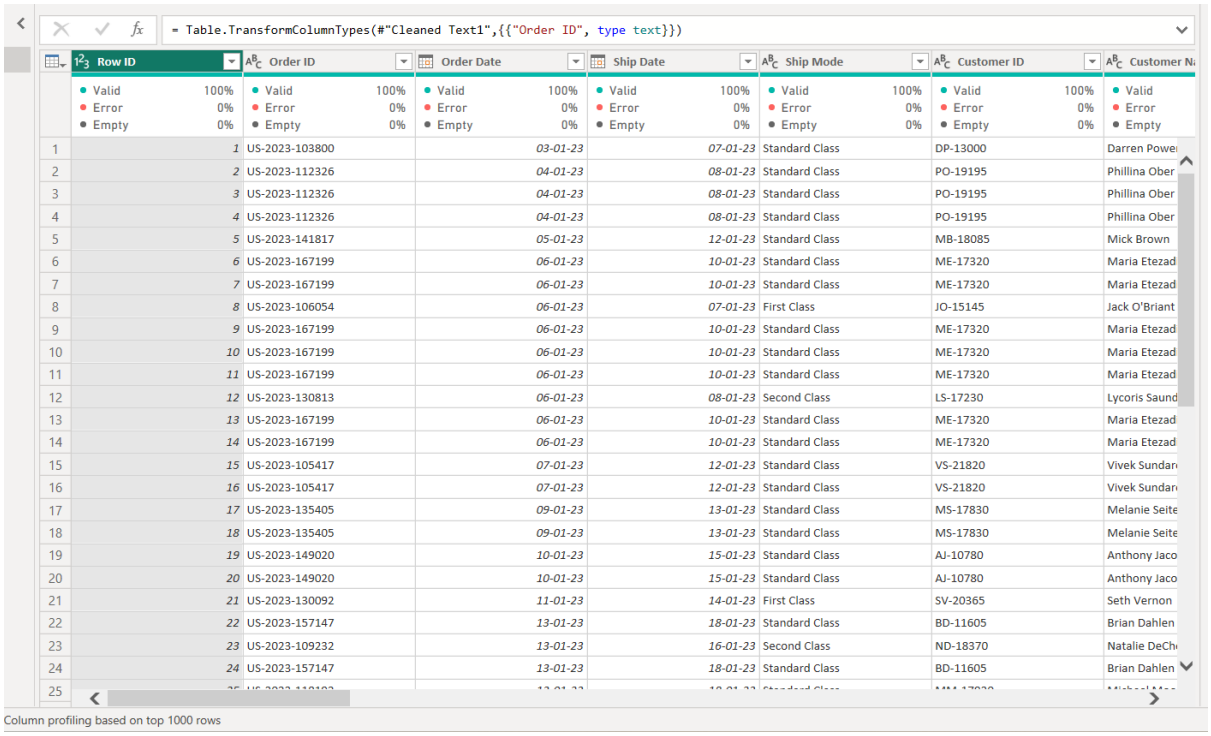
Trim and Clean transformations were applied to remove unnecessary spaces and unwanted characters.

Error Checking:

The columns were checked for:

- Errors
- Empty values
- Incorrect data types
- Formatting inconsistencies

After corrections, the data was verified to ensure that the transformation errors were removed.



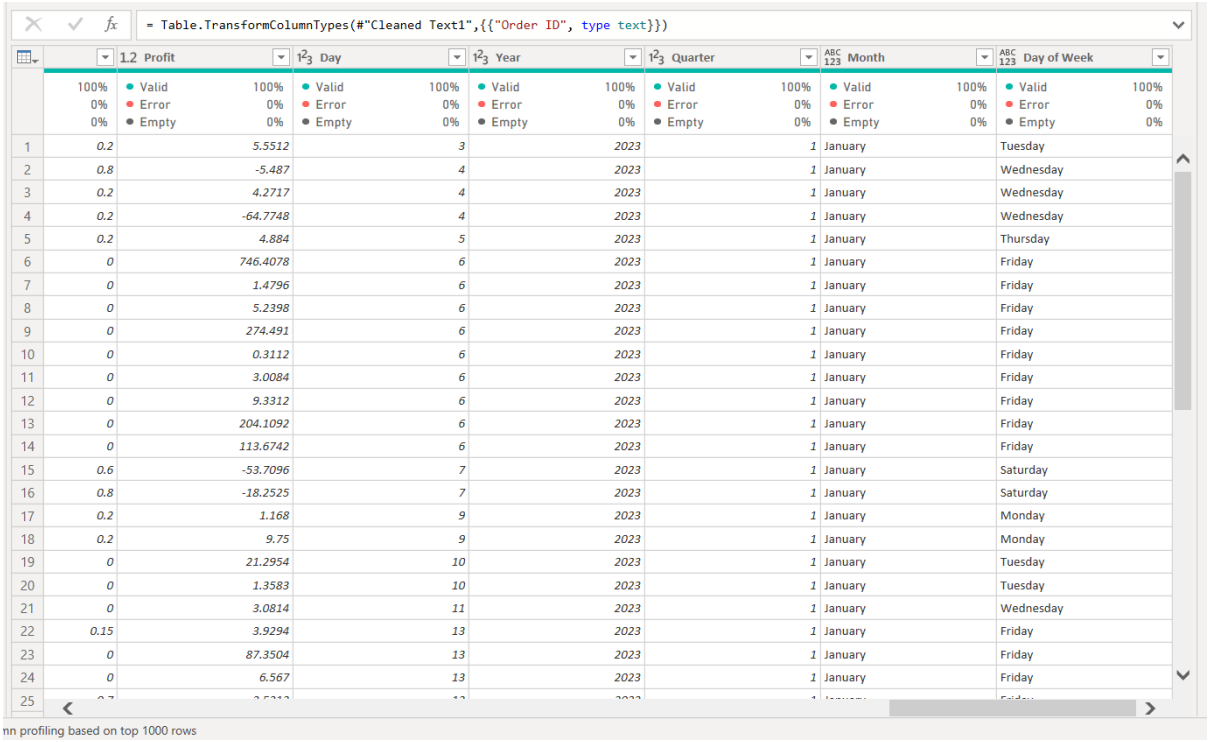
Date & Time Transformation:

The Order Date column was used to create additional temporal attributes.

The following columns were created:

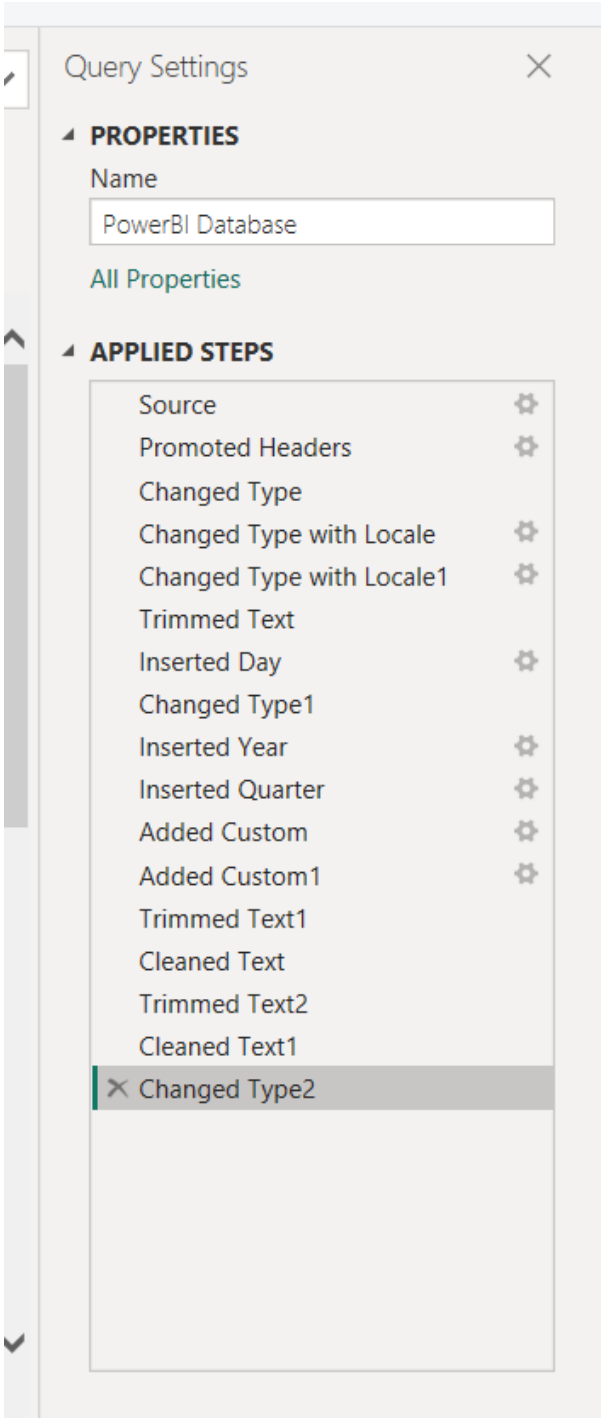
New Column	Purpose
Year	Identifies the year of the order
Quarter	Identifies the quarter
Month	Identifies the month
Day	Identifies the day of the month
Day of Week	Identifies the weekday

These fields will be useful for analyzing sales trends over time.



Power Query Applied Steps:

Power Query automatically maintains a sequence of Applied Steps, allowing the data-cleaning and transformation process to be tracked and reproduced.



Errors Encountered & Resolutions

This section will make your report look much more genuine and professional.

Error 1: Order ID Data Type

Problem:
The Order ID column was incorrectly assigned the Date data type, resulting in errors.

Solution:
The data type of Order ID was changed to Text.

Error 2: Incorrect Date Formula

Problem:
The Year and Quarter calculations initially referenced the wrong column.

Solution:
The formulas were corrected to use Order Date.

For example:

Year = Date.Year([Order Date])

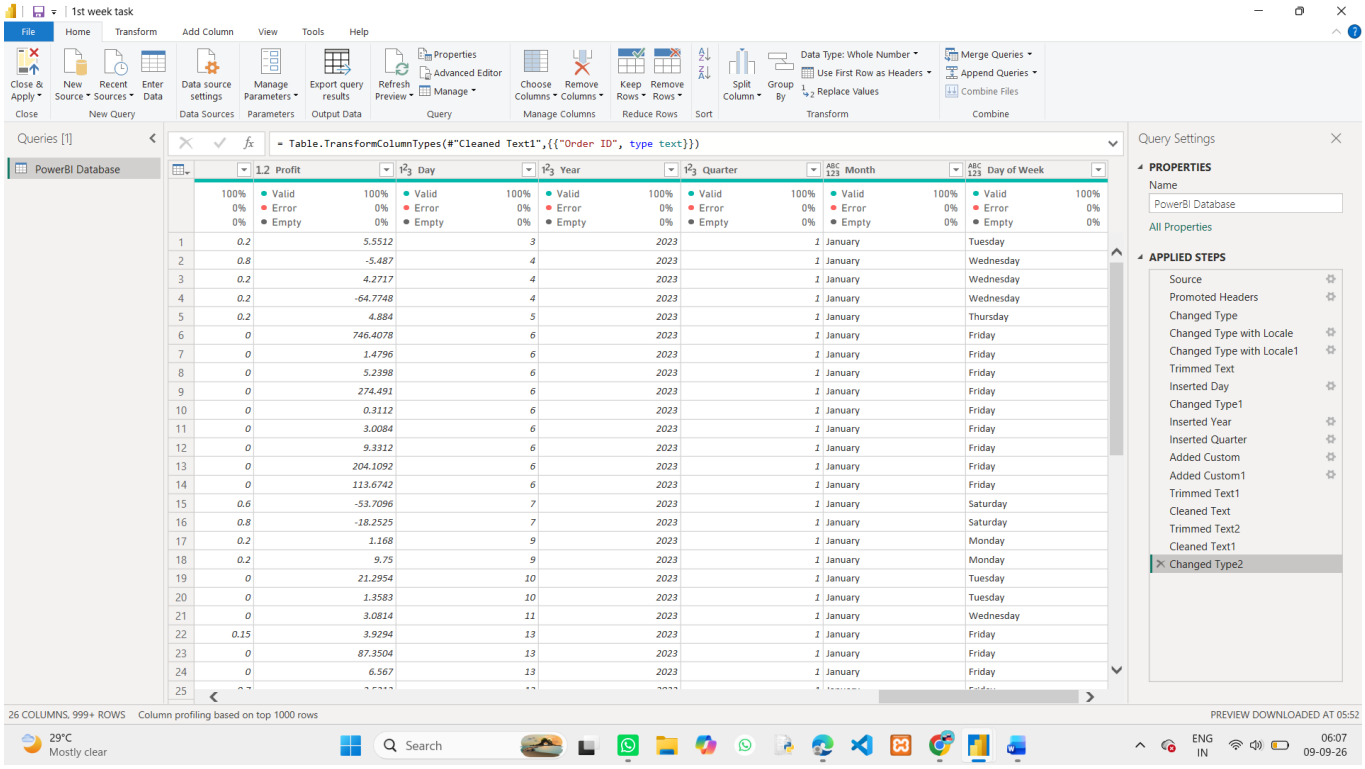
and

Quarter = Date.QuarterOfYear([Order Date])

After correction, the errors were removed.

Final Result:

After completing the data cleaning and transformation process, the dataset was successfully loaded into Power BI with corrected data types, standardized categorical fields, and additional date attributes. The dataset is now prepared for further analysis and visualization in the upcoming weeks.



Conclusion:

In Week 1, the Superstore Sales dataset was successfully imported and prepared using Power BI and Power Query. Data cleaning, categorical standardization, error correction, and date attribute extraction were completed. The resulting dataset provides a reliable foundation for creating interactive reports and dashboards in the subsequent stages of the project.